

Here is a **complete Spring Boot checklist** from **Beginner** → **Advanced** → **Expert level**, structured properly with **subtopics**, so you can track your preparation step-by-step.

Since you already have strong Java fundamentals (JDBC, Hibernate, Threads, Collections, etc.), I've structured this in a way that builds naturally on that knowledge.

1 Core Java & Spring Foundations (Pre-Spring Boot)

◆ **Java Prerequisites**

- OOP concepts
 - Exception handling
 - Collections Framework
 - Generics
 - Multithreading basics
 - JDBC
 - Reflection API
 - Annotations
-

◆ **Spring Core (Very Important)**

- What is Spring Framework?
- IoC (Inversion of Control)
- Dependency Injection
 - Constructor Injection
 - Setter Injection
- Bean Lifecycle
- Bean Scopes
 - Singleton
 - Prototype
 - Request
 - Session
- Spring Container
 - BeanFactory
 - ApplicationContext

- @Component, @Service, @Repository, @Controller
 - @Autowired
 - @Qualifier
 - @Primary
 - Java-based Configuration
 - XML Configuration (basic understanding)
-

2 Spring Boot Fundamentals (Beginner Level)

◆ Introduction

- What is Spring Boot?
 - Difference between Spring & Spring Boot
 - Why Spring Boot?
 - Spring Boot Architecture
-

◆ Project Setup

- Spring Initializr
 - Maven vs Gradle
 - Project structure
 - application.properties
 - application.yml
 - Profiles
-

◆ Auto Configuration

- @SpringBootApplication
 - Auto-configuration concept
 - @EnableAutoConfiguration
 - Component Scanning
-

◆ REST API Development

- @RestController
 - @RequestMapping
 - @GetMapping
 - @PostMapping
 - @PutMapping
 - @DeleteMapping
 - @PathVariable
 - @RequestParam
 - @RequestBody
 - ResponseEntity
-

◆ Configuration Properties

- @Value
 - @ConfigurationProperties
 - External configuration
-

3 Spring Boot with Database (Intermediate Level)

◆ Spring Data JPA

- What is Spring Data JPA?
 - JpaRepository
 - CrudRepository
 - PagingAndSortingRepository
 - Derived query methods
 - @Query annotation (JPQL & Native SQL)
 - Pagination & Sorting
 - Projections
-

◆ Hibernate Integration

- JPA vs Hibernate
 - Entity mapping
 - @Entity
 - @Table
 - @Id
 - @GeneratedValue
 - @OneToOne
 - @OneToMany
 - @ManyToOne
 - @ManyToMany
 - Fetch types
 - Cascade types
 - Lazy vs Eager loading
-

◆ Transactions

- @Transactional
 - Propagation types
 - Isolation levels
 - Rollback rules
-

◆ Database Migration

- Flyway
 - Liquibase
-

4 Validation & Exception Handling

◆ Validation

- @Valid
- @NotNull
- @Size
- @Email
- Custom validators

◆ Exception Handling

- @ControllerAdvice
 - @ExceptionHandler
 - Global exception handling
 - Custom exceptions
 - Response status handling
-

5 Spring Boot Security (Advanced)

◆ Spring Security Basics

- What is Spring Security?
 - Authentication vs Authorization
 - Security Filter Chain
 - Password encoding
 - BCrypt
-

◆ JWT Authentication

- JWT structure
 - Generating tokens
 - Validating tokens
 - Custom authentication filter
-

◆ Role-Based Authorization

- @PreAuthorize
 - Method-level security
 - Access control rules
-

◆ OAuth2

- OAuth2 basics
 - Social login
 - Authorization server vs Resource server
-

6 Advanced Spring Boot

◆ Caching

- @EnableCaching
 - @Cacheable
 - @CachePut
 - @CacheEvict
 - Redis integration
-

◆ Logging

- SLF4J
 - Logback configuration
 - Logging levels
-

◆ Scheduling

- @EnableScheduling
 - @Scheduled
-

◆ Asynchronous Processing

- @EnableAsync
- @Async
- CompletableFuture

◆ **Actuator**

- What is Spring Boot Actuator?
 - Health endpoints
 - Metrics
 - Custom endpoints
-

7 Microservices with Spring Boot

◆ **Microservices Basics**

- Monolith vs Microservices
 - Service communication
 - REST vs Messaging
-

◆ **Spring Cloud**

- What is Spring Cloud?
 - Service Discovery
 - Config Server
 - API Gateway
 - Circuit Breaker
-

◆ **API Gateway**

- Routing
 - Filters
 - Rate limiting
-

◆ **Service Discovery**

- Eureka basics
 - Registering services
-

◆ Inter-Service Communication

- RestTemplate
 - WebClient
 - OpenFeign
-

8 Messaging & Event-Driven Architecture

- Kafka integration
 - RabbitMQ integration
 - Producer & Consumer setup
 - Event-driven design
 - Idempotency
-

9 Testing in Spring Boot

◆ Unit Testing

- JUnit 5
 - Mockito
 - @MockBean
-

◆ Integration Testing

- @SpringBootTest
- @WebMvcTest
- TestRestTemplate

- H2 Database
-

Deployment & DevOps

- Packaging (Jar vs War)
 - Embedded Tomcat
 - Dockerizing Spring Boot app
 - CI/CD basics
 - AWS deployment basics
-

Performance & Production Readiness

- Connection pooling (HikariCP)
 - Thread pool tuning
 - Memory tuning
 - JVM parameters
 - Load balancing
 - Rate limiting
 - Monitoring with Prometheus & Grafana
-

If You Want Interview-Ready Mastery

Focus heavily on:

- Spring Core internals
- Spring Boot auto-configuration
- JPA performance tuning
- Transaction management
- Security (JWT)
- Microservices communication
- Exception handling best practices
- Logging + Monitoring

- Real-world project architecture
-

If you want, I can now:

- 🔥 Create a **3-month structured roadmap**
- 🔥 Give you **project ideas from beginner → advanced**
- 🔥 Create **interview Q&A checklist**
- 🔥 Or give you a **“must know 100 Spring Boot interview questions” list**

Tell me your goal (job switch? backend mastery? microservices?) and I'll tailor it exactly for you.